HW#1: Counting Invertible Matrices

Notation. We denote the space of $m \times m$ matrices over a finite field $\mathbb{F}_q$ by $\mathrm{Mat}_m(\mathbb{F}_q)$ instead of $\mathbb{F}_q^{m\times m}$.

Cardinality of the matrix space $\mathrm{Mat}_m(\mathbb{F}_q)$

Proposition 1. *Let $m, n \in \mathbb{Z}_{>0}$, and let q be a prime power. The number of $m \times n$ matrices over the finite field $\mathbb{F}_q$ is*

$$\left|\mathrm{Mat}_m(\mathbb{F}_q)\right| = q^{mn}.$$

In particular,

$$\left|\mathrm{Mat}_m(\mathbb{F}_q)\right| = q^{m^2}.$$

General linear group

Definition 1. The **general linear group** over $\mathbb{F}_q$ is

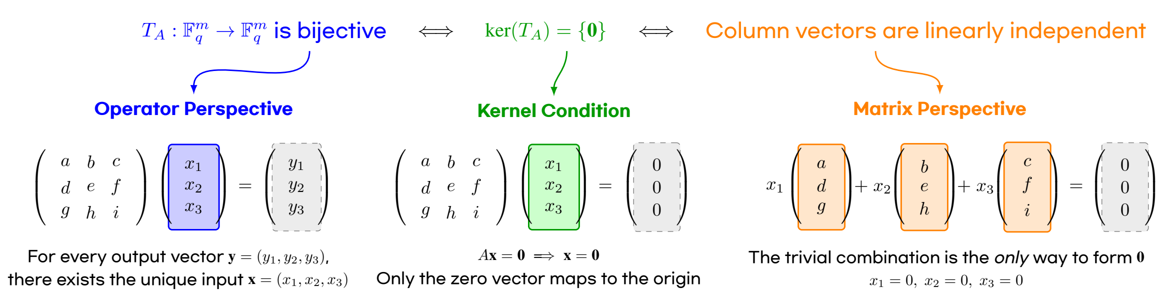
$$\mathrm{GL}_m(\mathbb{F}_q) := \{A \in \mathrm{Mat}_{m\times m}(\mathbb{F}_q) : A \text{ is invertible}\}.$$

Remark 1. $\mathrm{GL}_m(\mathbb{F}_q) = \{\, A \in \mathrm{Mat}_{m\times m}(\mathbb{F}_q) : \det A \neq 0 \,\} = \mathrm{Mat}_{m\times m}(\mathbb{F}_q) \quad \{\, A \in \mathrm{Mat}_{m\times m}(\mathbb{F}_q) : \det A = 0 \,\}.$

Remark 2. Let

$$A = \begin{pmatrix} | & | & & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_m \\ | & | & & | \end{pmatrix} \in \mathrm{Mat}_{m\times m}(\mathbb{F}_q),$$

where $\mathbf{v}_i \in \mathbb{F}_q^m$ are the columns of A. Then A is invertible if and only if its columns are linearly independent.



Obervation (Computation of $|GL_1(\mathbb{F}_q)|$).

A 1×1 matrix has the form

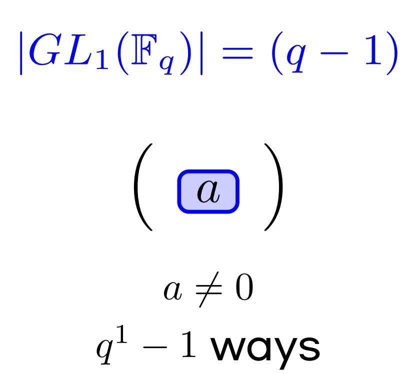
$$A = \big(a\big) = a, \qquad a \in \mathbb{F}_q.$$

Such a matrix is invertible iff $a \neq 0$. Therefore

$$GL_1(\mathbb{F}_q) = \left\{a : a \in \mathbb{F}_q^\times\right\},$$

and hence

$$|GL_1(\mathbb{F}_q)| = |\mathbb{F}_q^\times| = q - 1.$$



Obervation (Computation of $|GL_2(\mathbb{F}_q)|$).

A 2×2 matrix has the form

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad a, b, c, d \in \mathbb{F}_q.$$

Write the columns as

$$\mathbf{v}_1 = \begin{pmatrix} a \\ c \end{pmatrix}, \qquad \mathbf{v}_2 = \begin{pmatrix} b \\ d \end{pmatrix}.$$

$$|GL_2(\mathbb{F}_q)| = (q^2 - 1) \cdot (q^2 - q)$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$\mathbf{v}_1 \neq \mathbf{0}$ | $q^2 - 1$ ways $\times$ $\mathbf{v}_2 \notin \mathrm{Span}(\mathbf{v}_1)$ | $q^2 - q$ ways

(i) The vector $\mathbf{v}_1$ can be any nonzero vector in $\mathbb{F}_q^2$. Hence there are $q^2 - 1$ choices.

(ii) Since $\mathrm{Span}(\mathbf{v}_1) = \{ \lambda \mathbf{v}_1 : \lambda \in \mathbb{F}_q \}$ has exactly q elements, there are $q^2 - q$ choices for $\mathbf{v}_2$.

Multiplying these choices, we obtain

$$|GL_2(\mathbb{F}_q)| = (q^2 - 1)(q^2 - q).$$

Obervation (Computation of $|GL_3(\mathbb{F}_q)|$).

$$|GL_3(\mathbb{F}_q)| = (q^3 - 1) \cdot (q^3 - q) \cdot (q^3 - q^2)$$

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \qquad \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \qquad \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$$

$\mathbf{v}_1 \neq \mathbf{0}$ | $q^3 - 1$ ways $\times$ $\mathbf{v}_2 \notin \mathrm{Span}(\mathbf{v}_1)$ | $q^3 - q$ ways $\times$ $\mathbf{v}_3 \notin \mathrm{Span}(\mathbf{v}_1, \mathbf{v}_2)$ | $q^3 - q^2$ ways

Obervation (Computation of $|GL_4(\mathbb{F}_q)|$).

$$|GL_4(\mathbb{F}_q)| = (q^4 - 1) \cdot (q^4 - q) \cdot (q^4 - q^2) \cdot (q^4 - q^3)$$

$$\begin{pmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{pmatrix} \qquad \begin{pmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{pmatrix} \qquad \begin{pmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{pmatrix} \qquad \begin{pmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{pmatrix}$$

$\mathbf{v}_1 \neq \mathbf{0}$ | $q^4 - 1$ ways $\times$ $\mathbf{v}_2 \notin \mathrm{Span}(\mathbf{v}_1)$ | $q^4 - q$ ways $\times$ $\mathbf{v}_3 \notin \mathrm{Span}(\mathbf{v}_1, \mathbf{v}_2)$ | $q^4 - q^2$ ways $\times$ $\mathbf{v}_4 \notin \mathrm{Span}(\mathbf{v}_i)_{i=1}^{3}$ | $q^4 - q^3$ ways

Counting invertible matrices over a finite field $\mathbb{F}_q$

Theorem 2. *Let $m, n \in \mathbb{Z}_{>0}$, and let q be a prime power. The number of invertible $m \times m$ matrices over $\mathbb{F}_q$ is*

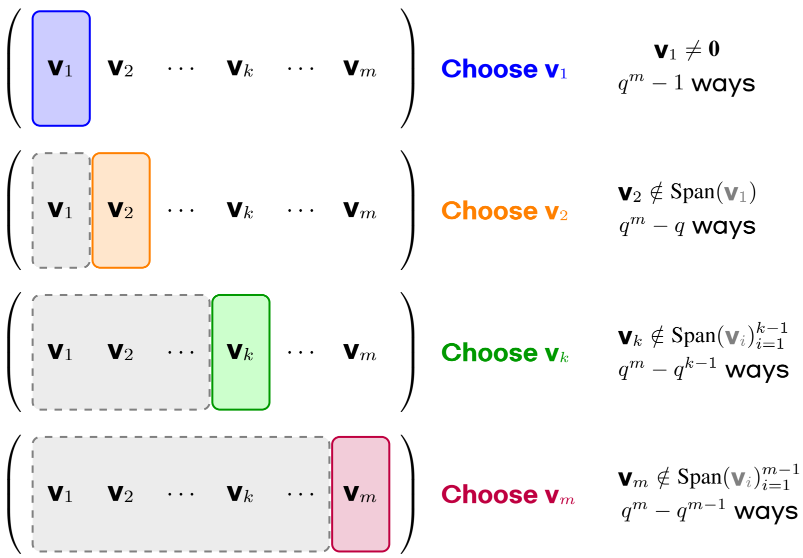
$$|\,\mathrm{GL}_m(\mathbb{F}_q)| = (q^m - 1)(q^m - q)(q^m - q^2)\cdots(q^m - q^{m-1}).$$

Equivalently,

$$|\,\mathrm{GL}_m(\mathbb{F}_q)| = \prod_{i=0}^{m-1}(q^m - q^i).$$

Remark 3.

$$|GL_m(\mathbb{F}_q)| = (q^m - 1)\cdot(q^m - q)\cdot\cdots\cdot(q^m - q^{k-1})\cdot\cdots\cdot(q^m - q^{m-1})$$



Remark 4. Recall that

$$|\,\mathrm{Mat}_m(\mathbb{F}_q)| = q^{m^2}, \qquad |GL_m(\mathbb{F}_q)| = \prod_{i=0}^{m-1}(q^m - q^i),$$

and hence

$$\begin{aligned}
\frac{|GL_m(\mathbb{F}_q)|}{|\,\mathrm{Mat}_m(\mathbb{F}_q)|} = \frac{\prod_{i=0}^{m-1}(q^m - q^i)}{q^{m^2}} &= \frac{1}{q^{m^2}}\left[(q^m - 1)(q^m - q)(q^m - q^2)\cdots(q^m - q^{m-1})\right] \\
&= \frac{1}{q^{m^2}}\left[q^m\left(1 - \frac{1}{q^m}\right)\left(1 - \frac{1}{q^{m-1}}\right)\left(1 - \frac{1}{q^{m-2}}\right)\cdots\left(1 - \frac{1}{q^1}\right)\right] \\
&= \left(1 - \frac{1}{q^m}\right)\left(1 - \frac{1}{q^{m-1}}\right)\left(1 - \frac{1}{q^{m-2}}\right)\cdots\left(1 - \frac{1}{q^1}\right) \\
&= \prod_{j=1}^{m}(1 - q^{-j}).
\end{aligned}$$

m	$\lvert\mathrm{Mat}_m(\mathbb{F}_q)\rvert$	$\lvert GL_m(\mathbb{F}_q)\rvert$	$\dfrac{\lvert GL_m(\mathbb{F}_q)\rvert}{\lvert\mathrm{Mat}_m(\mathbb{F}_q)\rvert}$
1	q	$q-1$	$1-q^{-1}$
2	q^4	$(q^2-1)(q^2-q)$	$(1-q^{-1})(1-q^{-2})$
3	q^9	$(q^3-1)(q^3-q)(q^3-q^2)$	$(1-q^{-1})(1-q^{-2})(1-q^{-3})$
4	q^{16}	$(q^4-1)(q^4-q)(q^4-q^2)(q^4-q^3)$	$(1-q^{-1})(1-q^{-2})(1-q^{-3})(1-q^{-4})$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
m	q^{m^2}	$\prod_{i=0}^{m-1}(q^m-q^i)$	$\prod_{j=1}^{m}(1-q^{-j})$

Example 1 (Invertible binary matrices)**.** For $q=2$, we have

$$|\mathrm{Mat}_m(\mathbb{F}_2)| = 2^{m^2}, \qquad \frac{|GL_m(\mathbb{F}_2)|}{|\mathrm{Mat}_m(\mathbb{F}_2)|} = \prod_{j=1}^{m}(1-2^{-j}).$$

m	$\lvert\mathrm{Mat}_m(\mathbb{F}_2)\rvert$	$\lvert GL_m(\mathbb{F}_2)\rvert$	$\dfrac{\lvert GL_m(\mathbb{F}_2)\rvert}{\lvert\mathrm{Mat}_m(\mathbb{F}_2)\rvert}$
1	2	1	0.500000
2	16	6	0.375000
3	512	168	0.328125
4	65, 536	20, 160	0.307617
$\vdots$	$\vdots$	$\vdots$	$\vdots$
20	–	–	0.288788

```
q = 2

print(f"{'m':>2} {'|Mat_m(F_q)|':>15} {'|GL_m(F_q)|':>15} {'ratio':>12}")
print("-" * 48)

for m in range(1, 7):
    Mat = q^(m^2)
    GL = prod(q^m - q^i for i in range(m))
    ratio = GL / Mat
    print(f"{m:>2} {Mat:>15} {GL:>15} {ratio.n():>12.6f}")
```

```
m    |Mat_m(F_q)|     |GL_m(F_q)|        ratio
------------------------------------------------
1               2               1     0.500000
2              16               6     0.375000
3             512             168     0.328125
4           65536           20160     0.307617
5        33554432         9999360     0.298004
6     68719476736     20158709760     0.293348
```

Proposition 3. *Let $q \geq 2$ be fixed, and define*

$$a_m := \frac{|GL_m(\mathbb{F}_q)|}{|\mathrm{Mat}_m(\mathbb{F}_q)|} = \prod_{j=1}^{m}(1 - q^{-j}).$$

Then the sequence $\langle a_m \rangle_{m \geq 1}$ converges as $m \to \infty$. In other words, there exists $\lim_{n\to\infty} \prod_{j=1}^{m}(1 - q^{-j}) \in \mathbb{R}$.

Proof. For every $j \geq 1$,

$$q \geq 2 \implies q^j \geq 2^j > 1 \implies 0 < \frac{1}{q^j} < 1.$$

Therefore

$$0 < \frac{1}{q^j} < 1 \implies -1 < -\frac{1}{q^j} < 0 \implies 0 < 1 - \frac{1}{q^j} < 1.$$

It follows that for each $m \geq 1$,

$$a_m = \prod_{j=1}^{m}(1 - q^{-j}) > 0.$$

So the sequence $\langle a_m \rangle$ is bounded below by 0. Moreover,

$$a_{m+1} = \prod_{j=1}^{m+1}(1 - q^{-j}) = \left[\prod_{j=1}^{m}(1 - q^{-j})\right] \cdot (1 - q^{-(m+1)}) = a_m(1 - q^{-(m+1)}).$$

Since $0 < 1 - q^{-(m+1)} < 1$, we obtain $a_{m+1} < a_m$. Thus $\langle a_m \rangle$ is decreasing. Therefore $\langle a_m \rangle$ is a monotone decreasing sequence which is bounded below. Hence it converges. $\square$

Example 2. For small prime q, the limit is approximately

$$q = 2: \qquad \prod_{j=1}^{\infty}(1 - 2^{-j}) \approx 0.288788095$$

$$q = 3: \qquad \prod_{j=1}^{\infty}(1 - 3^{-j}) \approx 0.560126077$$

$$q = 5: \qquad \prod_{j=1}^{\infty}(1 - 5^{-j}) \approx 0.760332795$$

```
q = 2
r = 3
s = 5
R = RealField(100)

L = prod(R(1) - R(q)^(-j) for j in range(1, 100))
M = prod(R(1) - R(r)^(-j) for j in range(1, 100))
N = prod(R(1) - R(s)^(-j) for j in range(1, 100))

print(L)
print(M)
print(N)
```

```
0.28878809508660242127889972193
0.56012607792794894496979224331
0.76033279587123242010148829629
```